PHC223 · Fall 2026

Pharmaceutical Analytical Chemistry 3

Question bank — 133 multiple-choice questions across 7 lectures. Answer key with explanations at the end.

Lecture 1 — Introduction to redox reactions

1. What is the equivalent weight of $KMnO_4$ (M = 158) in acid medium?
A. 158
B. 52.67
C. 31.6
D. 79

2. And in alkaline medium (forming MnO_2)?
A. 31.6
B. 52.67
C. 39.5
D. 158

3. An oxidizing agent:
A. Loses electrons
B. Has a low oxidation number
C. Is oxidized
D. Gains electrons and its oxidation number decreases

4. In the balanced reaction of MnO_4^- with Fe^{2+} in acid, the mole ratio MnO_4^- : Fe^{2+} is:
A. 1 : 1
B. 1 : 3
C. 1 : 5
D. 2 : 5

5. How many H^+ appear in the balanced dichromate–ferrous equation?
A. 6
B. 8
C. 14
D. 7

6. Oxidation is defined as:
A. Gain of electrons
B. Gain of hydrogen
C. Loss of electrons
D. Loss of oxygen

7. An oxidizing agent:
A. Loses electrons and is oxidized
B. Is never consumed
C. Gains protons only
D. Gains electrons and is reduced

8. In a redox reaction the number of electrons lost by the reductant is:
A. Greater than those gained
B. Unrelated
C. Smaller
D. Equal to those gained by the oxidant

9. The oxidation number of Mn in $KMnO_4$ is:
A. +2
B. +4
C. +6
D. +7

10. The oxidation number of Cr in $K_2Cr_2O_7$ is:

A. +3
B. +6
C. +7
D. +12

11. The oxidation number of S in $S_2O_3^{2-}$ (thiosulfate) is (average):

A. +2
B. +2.5
C. +4
D. +6

12. The oxidation number of S in $S_4O_6^{2-}$ (tetrathionate) is (average):

A. +2
B. +2.5
C. +4
D. +6

13. In acid medium MnO_4^- is reduced to:

A. MnO_2
B. Mn^{2+}
C. MnO_4^{2-}
D. Mn

14. In neutral or weakly alkaline medium MnO_4^- is reduced to:

A. MnO_2
B. Mn^{2+}
C. Mn^{3+}
D. Mn

15. The number of electrons gained when $Cr_2O_7^{2-} \rightarrow 2Cr^{3+}$ is:

A. 3
B. 5
C. 6
D. 7

16. Equivalent weight of $KMnO_4$ in acid medium is:

A. M/1
B. M/3
C. M/7
D. M/5

17. Equivalent weight of $K_2Cr_2O_7$ is:

A. M/2
B. M/6
C. M/5
D. M/3

18. In $2Fe^{3+} + Sn^{2+} \rightarrow 2Fe^{2+} + Sn^{4+}$, the reducing agent is:

A. Fe^{3+}
B. Sn^{4+}
C. Fe^{2+}
D. Sn^{2+}

19. H_2O_2 can act as:

A. Only an oxidant
B. Only a reductant
C. Neither
D. Both oxidant and reductant

20. When balancing a redox equation in acid, oxygen atoms are balanced by adding:
A. H_2O
B. OH^-
C. H^+ only
D. e^-

21. A substance whose oxidation number increases has been:
A. Oxidized
B. Reduced
C. Neutralized
D. Precipitated

Lecture 2 — Electrode potential

22. When solution pressure is greater than ionic pressure, the metal electrode:
A. Becomes positive and gains weight
B. Becomes negative and loses weight
C. Is unaffected
D. Plates out ions

23. According to $E = E^o + (0.059/n) \log[M^{n+}]$, E equals E^o when:
A. $[M^{n+}] = 1$ M
B. $[M^{n+}] = 0$
C. n = 1
D. The temperature is 0 °C

24. Electrode potential arises from:
A. Ion pairing
B. Gravity
C. A tendency of a metal to lose or gain electrons at the metal/solution interface
D. Light absorption

25. The standard hydrogen electrode has a potential of:
A. 1.00 V
B. 0.00 V by convention
C. −0.76 V
D. +0.34 V

26. Standard conditions for E° are:
A. 0 °C, 1 M, 1 atm
B. 25 °C, unit activity, 1 atm
C. 100 °C, 1 M
D. 25 °C, 0.1 M

27. The SHE uses which metal?
A. Platinized platinum
B. Silver
C. Copper
D. Mercury

28. A more positive reduction potential means the species is:
A. A stronger reductant
B. A stronger oxidant
C. Inert
D. A base

29. Zinc has E° = −0.76 V. This means zinc:
A. Is easily reduced
B. Does not react
C. Is a strong oxidant
D. Readily loses electrons (good reductant)

30. In a galvanic cell, oxidation occurs at the:
A. Cathode
B. Anode
C. Salt bridge
D. Both electrodes

31. Cell EMF is calculated as:
A. E(cathode) − E(anode)
B. E(anode) − E(cathode)
C. E(anode) + E(cathode)
D. E(cathode) × E(anode)

32. A reaction is spontaneous when E(cell) is:
A. Negative
B. Undefined
C. Zero
D. Positive

33. The salt bridge function is to:
A. Supply electrons
B. Maintain electrical neutrality between half-cells
C. Act as a catalyst
D. Measure voltage

34. In the Nernst equation at 25 °C, the factor 2.303RT/F equals about:
A. 0.0591 V
B. 0.591 V
C. 1.00 V
D. 0.00591 V

35. For Fe^{3+}/Fe^{2+}, increasing $[Fe^{3+}]/[Fe^{2+}]$ tenfold changes E by:
A. +0.0591 V
B. −0.0591 V
C. +0.591 V
D. 0

36. When [Ox] = [Red], the electrode potential equals:
A. E°
B. Zero
C. 2E°
D. 0.0591

37. The saturated calomel electrode contains:
A. Ag/AgCl
B. Hg/Hg_2Cl_2 in saturated KCl
C. Pt/H_2
D. $Cu/CuSO_4$

38. For a metal electrode M/M^{n+}, the Nernst equation has [Red] = 1 because:
A. The metal is a pure solid
B. The metal is a gas
C. Concentration is zero
D. It is the reference

Lecture 3 — Oxidation potential, SOP series and titration curves

39. The main role of a salt bridge is to:
A. Complete the circuit and keep the half-cells electrically neutral
B. Supply electrons
C. Raise the EMF
D. Act as the cathode

40. The standard oxidation potential (SOP) is the e.m.f. produced when:
A. A metal is dipped in its ion solution
B. A Pt electrode dipped in equal concentrations of Ox and Red forms is connected to the NHE
C. Two metals touch
D. A pH electrode is used

41. In the SOP series, the higher the SOP of a system:
A. The weaker its oxidized form as oxidant
B. No effect
C. The stronger its reduced form as reductant
D. The stronger the oxidizing power of its oxidized form and the weaker its reduced form as reductant

42. An oxidant can oxidize the reduced form of a system lying:
A. Above it in strength (stronger oxidant)
B. Below it (weaker oxidant system)
C. Anywhere
D. Only at equal potential

43. For a redox titration to be feasible, the potential difference between the two systems should be at least about:
A. 0.01 V
B. 0.05 V
C. 0.4 V
D. 5 V

44. According to the SOP series, Fe^{3+} can oxidize:
A. Cl^-
B. Br^-
C. All halides
D. I^- only

45. Oxidation potential differs from electrode potential in that it is measured with:
A. An inert Pt electrode in a solution of a redox system [Ox]/[Red]
B. A metal in its own ion solution
C. A glass electrode
D. No electrode

46. The redox titration curve plots:
A. Potential (E) vs volume of titrant
B. pH vs volume
C. Absorbance vs λ
D. Conductance vs time

47. Before the equivalence point in titrating Fe^{2+} with Ce^{4+}, E is calculated from:
A. Ce^{4+}/Ce^{3+} couple
B. Fe^{3+}/Fe^{2+} couple
C. Average of both
D. pH

48. At 50% titration of Fe^{2+}, E equals:
A. E° of Ce
B. E° of Fe^{3+}/Fe^{2+}
C. Zero
D. The equivalence potential

49. After the equivalence point, E is calculated from:
A. Titrant couple
B. Analyte couple
C. Indicator
D. Water

50. At the equivalence point when $n_1 = n_2 = 1$, E(eq) is:
A. $E_1° + E_2°$
B. $E_1° - E_2°$
C. $(E_1° + E_2°)/2$
D. Zero

51. General equivalence-point potential is:
A. $(E_1° + E_2°)/2$ always
B. $(n_1E_1° + n_2E_2°)/(n_1 + n_2)$
C. $n_1E_1° - n_2E_2°$
D. $E_1°·E_2°$

52. The redox titration curve is independent of dilution because:
A. E depends on volume
B. E depends on ratio of concentrations
C. Indicators fix it
D. Temperature is constant

53. A larger difference in E° between titrant and analyte gives:
A. A smaller jump
B. No end point
C. A larger potential break at equivalence
D. Negative E

54. When $n_1 \neq n_2$, the titration curve near equivalence is:
A. Symmetric
B. Asymmetric
C. Flat
D. Linear

55. The equilibrium constant of a redox reaction relates to ΔE° by:
A. $K = \Delta E°$
B. $\log K = n\Delta E°/0.0591$
C. $\log K = 0.0591/\Delta E°$
D. $K = n/\Delta E°$

56. At 200% titration (twice the equivalent volume), E equals:
A. E° of analyte
B. Equivalence potential
C. E° of titrant
D. Zero

Lecture 4 — Oxidation potential and the factors that change it

57. Adding a complexing agent that removes the oxidized form of a couple will:
A. Raise its potential
B. Not change it
C. Lower its potential
D. Double it

58. Fe^{3+} (E^o 0.77 V) can oxidize which halide?
A. Cl^-
B. Br^-
C. All halides
D. I^- only

59. In Zimmermann–Reinhardt reagent, the role of $MnSO_4$ is to:
A. Complex Fe^{3+}
B. Provide acidity
C. Act as indicator
D. Lower the MnO_4^-/Mn^{2+} potential below that of Cl_2/Cl^-

60. For a redox reaction to be complete, the two systems' potentials should differ by about:
A. 0.04 V
B. 0.1 V
C. 0.4 V
D. 1.5 V

61. For $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$, increasing acidity:
A. Decreases oxidizing power
B. Has no effect
C. Increases oxidizing power
D. Stops the reaction

62. Which factor does NOT change the oxidation potential of a system?
A. pH (when H^+ involved)
B. Complex formation
C. Colour of the container
D. Precipitation

63. Complexing the oxidized form of a couple:
A. Raises its oxidizing power
B. Makes it a stronger oxidant always
C. Has no effect
D. Lowers its oxidizing power (lowers E)

64. Adding phosphoric acid in Fe^{2+}/dichromate titration:
A. Complexes Fe^{3+} and lowers the Fe^{3+}/Fe^{2+} potential
B. Oxidizes Fe^{2+}
C. Acts as indicator
D. Raises pH

65. Fluoride added to Fe^{3+} solutions:
A. Increases E of Fe^{3+}/Fe^{2+}
B. Complexes Fe^{3+} (FeF_6^{3-}) and decreases E
C. Precipitates Fe^{2+}
D. Oxidizes iodide

66. Cu^{2+} normally cannot oxidize I^- by E°, yet it does because:
A. pH is high
B. Iodide is a strong oxidant
C. Light catalyzes it
D. CuI precipitates, lowering $[Cu^+]$ and raising E of Cu^{2+}/Cu^+

67. Precipitating the reduced form of a couple:
A. Decreases its potential
B. Has no effect
C. Increases its potential
D. Makes E zero

68. The arsenate/arsenite couple: H_3AsO_4 oxidizes I^- in strong acid, but As(III) is oxidized by I_2 in:
A. Neutral/weakly alkaline (bicarbonate) medium
B. Strongly acidic medium
C. Concentrated H_2SO_4
D. Any medium

69. Why is $NaHCO_3$ added when titrating As(III) with iodine?
A. To precipitate iodine
B. To colour the solution
C. To oxidize arsenite
D. To neutralize HI formed and drive the reaction forward

70. The formal potential differs from E° because it:
A. Ignores concentration
B. Is always zero
C. Uses activities of 1
D. Is measured at specified medium conditions (e.g. 1 M acid)

71. Increasing temperature in the Nernst equation:
A. Makes E° zero
B. Decreases it
C. Has no effect
D. Increases the slope factor RT/nF

72. For a couple not involving H^+, changing pH:
A. Strongly changes E
B. Doubles E
C. Does not affect E (unless hydrolysis/complexation)
D. Reverses E

73. Dichromate's oxidizing power increases with:
A. Increasing pH
B. Adding water only
C. Adding Cr^{3+}
D. Increasing $[H^+]$

74. An increase in the ratio [Red]/[Ox] causes E to:
A. Increase
B. Decrease
C. Stay the same
D. Become E°

Lecture 5 — Redox indicators, titrants and titration curves

75. Why is diphenylamine unsuitable for Fe^{2+} with dichromate unless phosphate is added?
A. Its E^o (0.76) is too close to Fe^{3+}/Fe^{2+} (0.77)
B. It is too strong
C. It reacts with Cl^-
D. It is a self-indicator

76. Which oxidizing titrant is a primary standard?
A. $KMnO_4$
B. Potassium dichromate
C. Ceric sulfate
D. Iodine

77. Ferroin changes colour at the end point from:
A. Colourless to violet
B. Yellow to green
C. Blue to red
D. Red to blue

78. 100 mL of 0.1 N KI is titrated with 0.1 N Ce^{4+} (E^o: I_2/I^- 0.54, n=2; Ce^{4+}/Ce^{3+} 1.44, n=1). Potential at equivalence?
A. 0.54 V
B. 0.84 V
C. 0.99 V
D. 1.44 V

79. Starch is classified as which type of redox indicator?
A. Self
B. External
C. Specific
D. Irreversible

80. Ferricyanide used on a spot plate to detect Fe^{2+} is an example of:
A. Self indicator
B. Specific adsorption indicator
C. Internal indicator
D. External indicator

81. Ferroin and diphenylamine are classified as:
A. Internal (redox) indicators
B. External indicators
C. Self indicators
D. pH indicators

82. Which titrant needs no added indicator?
A. $K_2Cr_2O_7$
B. $KMnO_4$ (and I_2)
C. Ce^{4+}
D. $Na_2S_2O_3$

83. A redox indicator changes colour when:
A. pH changes
B. Light is absent
C. Precipitate forms
D. It is oxidized or reduced at a characteristic potential

84. The colour-change range of a redox indicator is about:
A. $E°(In) \pm 1$ V
B. $E°(In) \pm 0.0591/n$ V
C. pKa ± 1
D. Exactly E°

85. $KMnO_4$ is described as a self-indicator because:
A. It is colourless
B. It fluoresces
C. It precipitates
D. Its intense purple colour persists after the end point

86. Diphenylamine changes colour to:
A. Red
B. Violet/blue on oxidation
C. Yellow
D. Green

87. Ferroin (1,10-phenanthroline–Fe(II)) changes from:
A. Colourless to pink
B. Blue to red
C. Red to pale blue on oxidation
D. Yellow to green

88. Starch is a specific indicator for:
A. Iodine (blue complex)
B. Fe^{3+}
C. MnO_4^-
D. Cr^{3+}

89. An indicator is suitable when its transition potential:
A. Falls within the steep part of the titration curve near equivalence
B. Is far from equivalence
C. Equals 0 V
D. Is negative

90. $KMnO_4$ is not a primary standard because:
A. It is too pure
B. It is colourless
C. It contains MnO_2 and decomposes in solution
D. It is insoluble

91. $KMnO_4$ is standardized against:
A. NaCl
B. Sodium oxalate
C. KI
D. Borax

92. Why is HCl avoided in permanganate titrations?
A. It is too weak
B. It reduces Fe^{3+}
C. It forms a precipitate
D. Cl^- is oxidized to Cl_2 by MnO_4^-

93. The oxalate–permanganate reaction is autocatalyzed by:
A. Fe^{2+}
B. Mn^{2+}
C. H^+
D. O_2

94. $K_2Cr_2O_7$ is preferred over $KMnO_4$ because it:
A. Is a stronger oxidant
B. Is cheaper always
C. Is a self-indicator
D. Is a primary standard and stable in solution

95. Cerium(IV) sulfate advantages include:
A. Purple colour
B. Single-electron change and stability
C. Only works in base
D. It is a reductant

96. Reductants such as Fe^{2+} must be protected from:
A. Light only
B. CO_2
C. Heat only
D. Air oxidation

97. The Zimmermann–Reinhardt reagent is used to:
A. Make starch
B. Titrate Fe^{2+} with $KMnO_4$ in presence of chloride
C. Titrate iodine
D. Remove oxygen

98. Before titrating total iron with dichromate, Fe^{3+} is reduced with:
A. Ce^{4+}
B. $KMnO_4$
C. H_2O_2
D. $SnCl_2$

Lecture 6 — Iodimetry, iodometry, KIO_3 and $KBrO_3$

99. H_2O_2 is determined by adding excess KI and titrating the liberated I_2 with thiosulfate. This is:
A. Iodometry
B. Iodimetry
C. Cerimetry
D. Permanganometry

100. Thiosulfate is oxidized by iodine to:
A. Tetrathionate
B. Sulfite
C. Sulfate
D. Sulfur

101. In strongly acidic medium, how many electrons does each IO_3^- gain?
A. 6
B. 5
C. 4
D. 2

102. Iodimetry is:
A. Direct titration with standard iodine solution
B. Titration of liberated iodine with thiosulfate
C. Titration with KIO_3 only
D. Precipitation titration

103. Iodometry is:
A. Direct titration with iodine
B. Titration with bromine
C. Indirect: oxidant liberates I_2 from excess KI, then I_2 titrated with thiosulfate
D. Titration with Ce^{4+}

104. Iodine is dissolved in KI solution because:
A. I_2 is very soluble in water
B. It oxidizes KI
C. KI is an indicator
D. It forms soluble I_3^- and reduces volatility

105. Thiosulfate reacts with iodine to give:
A. Tetrathionate
B. Sulfate
C. Sulfur
D. Sulfite

106. In iodometry, starch is added:
A. At the beginning
B. Never
C. Near the end point when the solution is pale yellow
D. With the KI

107. Iodometric titrations must not be performed in strongly alkaline medium because:
A. Thiosulfate is stable
B. Starch precipitates
C. Iodine disproportionates to iodide and hypoiodite/iodate
D. KI oxidizes

108. In strongly acidic medium thiosulfate:
A. Is stable
B. Decomposes to sulfur and SO_2
C. Turns blue
D. Precipitates iodine

109. Sodium thiosulfate solution can be standardized with:
A. Borax
B. NaCl
C. KIO_3 + excess KI in acid
D. Oxalic acid alone

110. $IO_3^- + 5I^- + 6H^+ \rightarrow ?\ I_2$
- A. 1
- B. 2
- C. 3
- D. 5

111. Ascorbic acid (vitamin C) can be assayed by:
- A. Permanganate in base
- B. Iodimetry
- C. Argentometry
- D. Complexometry

112. Cu^{2+} is determined iodometrically because:
- A. It forms CuI and liberates I_2 from KI
- B. It reduces I_2
- C. It forms a blue complex
- D. It's a primary standard

113. In Andrews' titration, KIO_3 reacts in strong HCl to form:
- A. I_2
- B. I^-
- C. ICl (iodine monochloride)
- D. IO_4^-

114. Equivalent weight of KIO_3 in Andrews' titration is:
- A. M/1
- B. M/6
- C. M/5
- D. M/4

115. Equivalent weight of KIO_3 when reacting with excess KI in dilute acid (to I_2) is:
- A. M/4
- B. M/6
- C. M/5
- D. M/2

116. $KBrO_3$ in acid with excess KBr generates:
- A. Br^-
- B. HBr only
- C. BrO_4^-
- D. Br_2

117. Bromine generated from $KBrO_3$ is used to determine phenol by:
- A. Complexation
- B. Oxidation to CO_2
- C. Bromination to tribromophenol
- D. Precipitation as AgBr

118. Equivalent weight of $KBrO_3$ ($BrO_3^- \rightarrow Br^-$) is:
- A. M/2
- B. M/6
- C. M/5
- D. M/3

119. Methyl orange end point in bromatometry occurs because:
- A. It reduces bromate
- B. It turns blue with iodine
- C. It precipitates
- D. First excess Br_2 destroys (bleaches) the dye

120. Iodine solutions are stored in:
A. Dark glass-stoppered bottles
B. Clear plastic
C. Open beakers
D. Metal cans

121. Thiosulfate solutions are stabilized by adding a little:
A. HCl
B. $KMnO_4$
C. Na_2CO_3 and using boiled water
D. Starch

Lecture 7 — Midterm revision

122. Which species is reduced in $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$?
A. Zn
B. Cu^{2+}
C. Zn^{2+}
D. Cu

123. 0.1 N $KMnO_4$ in acid corresponds to a molarity of:
A. 0.1 M
B. 0.02 M
C. 0.5 M
D. 0.05 M

124. 0.1 N $K_2Cr_2O_7$ corresponds to:
A. 0.0167 M
B. 0.1 M
C. 0.6 M
D. 0.05 M

125. 25 mL of 0.1 N Fe^{2+} requires how many mL of 0.1 N $KMnO_4$?
A. 5
B. 12.5
C. 25
D. 50

126. The Nernst equation for MnO_4^-/Mn^{2+} includes $[H^+]$ raised to the power:
A. 1
B. 5
C. 8
D. 14

127. Fe^{3+}/Fe^{2+} $E° = 0.77$ V; Ce^{4+}/Ce^{3+} $E° = 1.44$ V. E at equivalence is about:
A. 0.67 V
B. 1.105 V
C. 2.21 V
D. 0.77 V

128. Which is a primary standard oxidant?
A. $KMnO_4$
B. $Ce(SO_4)_2$ unstandardized
C. I_2
D. $K_2Cr_2O_7$

129. Which is a self-indicating titrant?
A. $K_2Cr_2O_7$
B. $Na_2S_2O_3$
C. $KMnO_4$
D. $KBrO_3$

130. In iodometric determination of an oxidant, the thiosulfate volume is proportional to:

A. Excess KI
B. Water volume
C. Starch amount
D. The amount of oxidant

131. A positive ΔE° between oxidant and reductant systems indicates:

A. No reaction
B. Spontaneous reaction
C. Reverse reaction only
D. Equilibrium constant < 1

132. Which pair is correct?

A. Iodometry – titrant thiosulfate
B. Iodimetry – titrant thiosulfate
C. Iodimetry – indirect
D. Iodometry – direct with I_2

133. Mn^{2+} added at the start of oxalate titration:

A. Speeds it (catalyst)
B. Slows the reaction
C. Is an indicator
D. Is the analyte

Answer key

Lecture 1

1	C	**31.6** — Mn goes from +7 to +2: a change of 5. 158 ÷ 5 = 31.6.
2	B	**52.67** — Mn goes from +7 to +4: a change of 3. 158 ÷ 3 = 52.67.
3	D	**Gains electrons and its oxidation number decreases** — The oxidant accepts electrons, so it is itself reduced.
4	C	**1 : 5** — MnO_4^- gains 5 electrons and each Fe^{2+} gives 1, so 1 : 5.
5	C	**14** — $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$.
6	C	**Loss of electrons** — OIL RIG: Oxidation Is Loss, Reduction Is Gain.
7	D	**Gains electrons and is reduced** — It takes electrons from the reductant.
8	D	**Equal to those gained by the oxidant** — Electrons are conserved.
9	D	**+7** — K(+1) + Mn + 4(−2) = 0 → Mn = +7.
10	B	**+6** — 2(+1) + 2Cr + 7(−2) = 0 → Cr = +6.
11	A	**+2** — 2S + 3(−2) = −2 → S = +2.
12	B	**+2.5** — 4S − 12 = −2 → S = +2.5.
13	B	**Mn^{2+}** — A 5-electron change.
14	A	**MnO_2** — A 3-electron change, brown precipitate.
15	C	**6** — Each Cr goes +6 → +3; two Cr atoms = 6 e^-.
16	D	**M/5** — Molar mass ÷ electrons transferred (5).
17	B	**M/6** — Six electrons per mole.
18	D	**Sn^{2+}** — Sn^{2+} loses 2 electrons.
19	D	**Both oxidant and reductant** — O is −1, between 0 and −2.
20	A	**H_2O** — Then H with H^+, then charge with e^-.
21	A	**Oxidized** — Loss of electrons raises oxidation number.

Lecture 2

22	B	**Becomes negative and loses weight** — The metal dissolves as ions, leaving electrons behind: negative charge, weight loss.
23	A	**$[M^{n+}] = 1$ M** — log 1 = 0, so E = E^o at unit ion concentration.
24	C	**A tendency of a metal to lose or gain electrons at the metal/solution interface** — An electrical double layer forms.
25	B	**0.00 V by convention** — The reference for all standard potentials.
26	B	**25 °C, unit activity, 1 atm** — Activities of 1.
27	A	**Platinized platinum** — Platinum black, H_2 at 1 atm, H^+ activity 1.
28	B	**A stronger oxidant** — It is more easily reduced.
29	D	**Readily loses electrons (good reductant)** — Negative E° = strong tendency to be oxidized.
30	B	**Anode** — An Ox, Red Cat.
31	A	**E(cathode) − E(anode)** — Using reduction potentials.
32	D	**Positive** — ΔG = −nFE, so positive E gives negative ΔG.
33	B	**Maintain electrical neutrality between half-cells** — Ions migrate to balance charge.
34	A	**0.0591 V** — E = E° + (0.0591/n) log([Ox]/[Red]).
35	A	**+0.0591 V** — n = 1, log 10 = 1.
36	A	**E°** — log 1 = 0.
37	B	**Hg/Hg_2Cl_2 in saturated KCl** — A common reference electrode (~+0.24 V).
38	A	**The metal is a pure solid** — Pure solids have unit activity.

Lecture 3

39	**A**	**Complete the circuit and keep the half-cells electrically neutral** — Ions migrate through the bridge, balancing the charge build-up in each half-cell.
40	**B**	**A Pt electrode dipped in equal concentrations of Ox and Red forms is connected to the NHE** — Inert electrode, [Ox] = [Red], versus NHE.
41	**D**	**The stronger the oxidizing power of its oxidized form and the weaker its reduced form as reductant** — Strong oxidants sit at the top (high +ve); strong reductants at the bottom.
42	**B**	**Below it (weaker oxidant system)** — A stronger oxidant oxidizes the reduced form of a weaker one.
43	**C**	**0.4 V** — The lecture states about 0.4 V between the two systems.
44	**D**	**I^- only** — MnO_4^- can oxidize Br^-, I^- and Fe^{2+} (the lecture example).
45	**A**	**An inert Pt electrode in a solution of a redox system [Ox]/[Red]** — Electrode potential uses a metal dipped in its ions.
46	**A**	**Potential (E) vs volume of titrant** — Analogous to a pH titration curve.
47	**B**	**Fe^{3+}/Fe^{2+} couple** — The analyte couple is present in measurable ratio.
48	**B**	**E° of Fe^{3+}/Fe^{2+}** — $[Fe^{3+}] = [Fe^{2+}]$.
49	**A**	**Titrant couple** — Excess titrant controls potential.
50	**C**	**$(E_1° + E_2°)/2$** — The arithmetic mean.
51	**B**	**$(n_1E_1° + n_2E_2°)/(n_1 + n_2)$** — Weighted by electrons.
52	**B**	**E depends on ratio of concentrations** — Ratios don't change on dilution (approximately).
53	**C**	**A larger potential break at equivalence** — Sharper end point.
54	**B**	**Asymmetric** — Equivalence point shifts toward the couple with larger n.
55	**B**	**$\log K = n\Delta E°/0.0591$** — Large $\Delta E°$ → very large K.
56	**C**	**E° of titrant** — $[Ce^{4+}] = [Ce^{3+}]$.

Lecture 4

57	**C**	**Lower its potential** — $E = E^o + (0.059/n) \log [Ox]/[Red]$. Less [Ox] → lower E.
58	**D**	**I^- only** — Only I_2/I^- (0.54 V) lies below Fe^{3+}/Fe^{2+}.
59	**D**	**Lower the MnO_4^-/Mn^{2+} potential below that of Cl_2/Cl^-** — Extra Mn^{2+} lowers the permanganate potential so chloride is no longer oxidized.
60	**C**	**0.4 V** — A difference of about 0.4 V is needed for quantitative reaction.
61	**C**	**Increases oxidizing power** — $[H^+]^8$ appears in the numerator of the Nernst term.
62	**C**	**Colour of the container** — pH, complexation, precipitation and concentration ratio matter.
63	**D**	**Lowers its oxidizing power (lowers E)** — It lowers [Ox] in the Nernst ratio.
64	**A**	**Complexes Fe^{3+} and lowers the Fe^{3+}/Fe^{2+} potential** — Sharpens the end point with diphenylamine.
65	**B**	**Complexes Fe^{3+} (FeF_6^{3-}) and decreases E** — Used to prevent Fe^{3+} from oxidizing I^-.
66	**D**	**CuI precipitates, lowering $[Cu^+]$ and raising E of Cu^{2+}/Cu^+** — Precipitation of the reduced form raises the potential.
67	**C**	**Increases its potential** — Lower [Red] in the Nernst ratio.
68	**A**	**Neutral/weakly alkaline (bicarbonate) medium** — pH reverses the reaction direction.
69	**D**	**To neutralize HI formed and drive the reaction forward** — Removing H^+ shifts equilibrium.
70	**D**	**Is measured at specified medium conditions (e.g. 1 M acid)** — Includes activity and side-reaction effects.
71	**D**	**Increases the slope factor RT/nF** — T is in the numerator.
72	**C**	**Does not affect E (unless hydrolysis/complexation)** — E.g. Fe^{3+}/Fe^{2+} in acid.
73	**D**	**Increasing $[H^+]$** — 14 H^+ consumed in the half-reaction.
74	**B**	**Decrease** — log term becomes negative.

Lecture 5

75	**A**	**Its E^o (0.76) is too close to Fe^{3+}/Fe^{2+} (0.77)** — Phosphate complexes Fe^{3+}, lowering the iron potential so the indicator changes after the end point.

76	B	**Potassium dichromate** — $K_2Cr_2O_7$ is a primary standard. $KMnO_4$ must be standardized against sodium oxalate or As_2O_3.
77	D	**Red to blue** — Ferroin (E^o 1.06): red (reduced) → blue (oxidized).
78	B	**0.84 V** — E = (2×0.54 + 1×1.44)/(2+1) = 2.52/3 = 0.84 V.
79	C	**Specific** — Starch reacts specifically with iodine (blue complex).
80	D	**External indicator** — Drops of the titrated solution are tested outside the flask.
81	A	**Internal (redox) indicators** — Added to the titration flask.
82	B	**$KMnO_4$ (and I_2)** — Self indicators.
83	D	**It is oxidized or reduced at a characteristic potential** — Its two forms have different colours.
84	B	**$E°(In) \pm 0.0591/n$ V** — Ratio from 10:1 to 1:10.
85	D	**Its intense purple colour persists after the end point** — Mn^{2+} is nearly colourless.
86	B	**Violet/blue on oxidation** — Used with dichromate.
87	C	**Red to pale blue on oxidation** — Used in cerimetry.
88	A	**Iodine (blue complex)** — Triiodide–amylose complex.
89	A	**Falls within the steep part of the titration curve near equivalence** — Like choosing a pH indicator.
90	C	**It contains MnO_2 and decomposes in solution** — Standardize against sodium oxalate.
91	B	**Sodium oxalate** — Heated to ~70 °C in H_2SO_4.
92	D	**Cl^- is oxidized to Cl_2 by MnO_4^-** — Use H_2SO_4 instead.
93	B	**Mn^{2+}** — Slow at first, then fast.
94	D	**Is a primary standard and stable in solution** — It can be used in HCl too.
95	B	**Single-electron change and stability** — $Ce^{4+} + e^- \rightarrow Ce^{3+}$; stable solutions.
96	D	**Air oxidation** — Prepared in acid and standardized frequently.
97	B	**Titrate Fe^{2+} with $KMnO_4$ in presence of chloride** — $MnSO_4 + H_3PO_4 + H_2SO_4$.
98	D	**$SnCl_2$** — Excess Sn^{2+} removed with $HgCl_2$.

Lecture 6

99	A	**Iodometry** — Indirect method for an oxidizing sample = iodometry.
100	A	**Tetrathionate** — $I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$.
101	C	**4** — $IO_3^- + 6H^+ + 4e \rightarrow I^+ + 3H_2O$. Neutral: 6 e; slightly acidic: 5 e.
102	A	**Direct titration with standard iodine solution** — Iodine acts as a mild oxidant.
103	C	**Indirect: oxidant liberates I_2 from excess KI, then I_2 titrated with thiosulfate** — Used for oxidizing analytes.
104	D	**It forms soluble I_3^- and reduces volatility** — $I_2 + I^- \rightleftharpoons I_3^-$.
105	A	**Tetrathionate** — $2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$.
106	C	**Near the end point when the solution is pale yellow** — High I_2 forms a stubborn complex.
107	C	**Iodine disproportionates to iodide and hypoiodite/iodate** — Also thiosulfate oxidizes to sulfate.
108	B	**Decomposes to sulfur and SO_2** — Also air oxidizes I^- in strong acid.
109	C	**KIO_3 + excess KI in acid** — KIO_3 is a primary standard.
110	C	**3** — $3I_2 + 3H_2O$.
111	B	**Iodimetry** — Iodine oxidizes it to dehydroascorbic acid.
112	A	**It forms CuI and liberates I_2 from KI** — $2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$.
113	C	**ICl (iodine monochloride)** — A 4-electron change; end point by disappearance of I_2 colour in CCl_4/chloroform.
114	D	**M/4** — I(+5) → I(+1).
115	B	**M/6** — 1 IO_3^- gives 3 I_2 = 6 equivalents of thiosulfate.
116	D	**Br_2** — $BrO_3^- + 5Br^- + 6H^+ \rightarrow 3Br_2 + 3H_2O$.
117	C	**Bromination to tribromophenol** — Excess Br_2 back-titrated iodometrically.
118	B	**M/6** — Br +5 → −1 = 6 e^-.

119	**D**	**First excess Br_2 destroys (bleaches) the dye** — An irreversible indicator.
120	**A**	**Dark glass-stoppered bottles** — Light and volatility; iodine attacks rubber.
121	**C**	**Na_2CO_3 and using boiled water** — Prevents bacterial and acid decomposition.

Lecture 7

122	**B**	**Cu^{2+}** — Cu^{2+} gains 2 e^-.
123	**B**	**0.02 M** — N = M × 5.
124	**A**	**0.0167 M** — N = M × 6.
125	**C**	**25** — $N_1V_1 = N_2V_2$.
126	**C**	**8** — 8 H^+ in the half-reaction.
127	**B**	**1.105 V** — (0.77 + 1.44)/2.
128	**D**	**$K_2Cr_2O_7$** — Also KIO_3, $KBrO_3$.
129	**C**	**$KMnO_4$** — Pink persists at end point.
130	**D**	**The amount of oxidant** — Liberated I_2 equals oxidant equivalents.
131	**B**	**Spontaneous reaction** — log K = nΔE°/0.0591.
132	**A**	**Iodometry – titrant thiosulfate** — Iodimetry uses standard I_2 directly.
133	**A**	**Speeds it (catalyst)** — Autocatalysis.